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Total No. of Questions : 13

Total No. of Pages : 02

B.Pharmacy (Sem.–6)

BIOPHARMACEUTICS AND PHARMACOKINETICS-THEORY

Subject Code : BP-604T

M.Code : 77989

Date of Examination : 10-06-23

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and students have to attempt any SEVEN questions.

SECTION-A

1. Write briefly :

   a) What do you mean by renal clearance?

   b) Name the two plasma proteins involved in drug-protein binding of basic drugs.

   c) What is absolute bioavailability?

   d) What do you mean by rate of drug excretion?

   e) What is MRT and what does it represent?

   f) What are therapeutic equivalents?

   g) What is active secretion in renal tubules?

   h) Mention the formula for calculating $t_{1/2}$ of drug for first order kinetics.

   i) What is pinocytosis?

   j) What is facilitated transport?

SECTION-B

2. What is bioequivalence? Mention the criteria for declaring two products bioequivalent and discuss the regulatory considerations pertaining to bioequivalence studies in India.

3. A dose of [illegible] mg was given to a patient by IV bolus injection. [illegible] drug concentration was found to be [illegible]. Assume first order kinetics. [illegible]

4. Discuss physicochemical factors influencing drug absorption after oral administration.

SECTION-C

5. How do the pKa of drug and the pH of the GI fluid influence the absorption of drugs?

6. Briefly discuss the in vitro dissolution methods.

7. Give a brief introduction of non-compartment model and discuss its difference with compartment model.

8. What is the drug concentration in plasma at [illegible] after [illegible] of a single dose of 10 mg. The [illegible]

9. Write briefly about renal clearance ratio.

10. Discuss the phase I reactions for drug metabolism.

11. Discuss the sigma-minus method to estimate elimination rate constant from urinary excretion data.

12. Discuss the Wagner-Nelson method with suitable equation.

13. Considering [illegible] steady state [illegible] loading dose [illegible]

NOTE: Disclosure of identity by writing mobile number or making passing request on any page of Answer sheet will lead to UFM against the Student.